

National Physics Olympiad: Category-C Finals

Marks: 20

March 15, 2019

You should clearly label your answers with the corresponding question numbers

1: Efficient Engineer

An engineer is given the task to extract the maximum work out of two bodies A and B whose heat capacities are C_A and C_B respectively. They are kept at temperatures T_A and T_B (in Kelvin scale). The engineer first tries to put these two bodies in contact.

(a) What would be the amount of work extracted during this process?

The engineer then tries to design an engine to get the best output (work) from this situation.

(b) What in your opinion would be the best method to achieve this goal?

(c) Find the value of the parameter for this "engine" that will guarantee that the work is extracted in the method is indeed maximum. **1+2+2=5**

2: Damping in Plasmas

Plasma is usually called the fourth state of matter. In a plasma, matter remains in ionized state but it remains electrically neutral as a whole. One can have waves inside the plasma just like waves in a fluid but these carry electric field with them. In the plasma, the restoration force is provided by the electrostatic interaction between the ions. These waves can lose their energy to the charged particles in the plasma as the charge particles experience resonance due to the interaction with the waves. As a consequence the charged particles are accelerated. This process is historically known as Landau damping. Let us try to see how this process can be understood in a crude way.

Consider a particle with charge q and velocity v_x (which is moving along x -axis). A transverse electric field (associated with the plasma wave and pointing in the z -direction) comes and "hits" the particle. Let this electric field be given by

$$E_z(x) = E_0 \cos(kx - \omega t)$$

Note the wave is propagating in the x -direction.

(a) What is the equation of motion for the charged particle?

(b) Find the velocity of the particle along the z -direction as a function of time.

(c) What is the kinetic energy associated with the motion along the z -direction which is now added to the particle, due to the incident electric wave?

(d) If $f(v_x)dv_x$ is the number of particles in the plasma with velocities lying within the range v_x and $v_x + dv_x$ what is the energy transferred per unit time from the wave to the particles? You can choose the distribution function $f(v_x)$ to vary slowly near the value $v_x = \frac{\omega}{k}$.

You are given $\int_{-\infty}^{\infty} \frac{\sin^2 u}{u^2} du = \pi$.

1+2+1+3=7

3: How bright can stars be?

Stars are formed by the accretion or "collapse" of galactic dust under their own gravity and in that process heating up and consequently starts thermonuclear reactions. When there is a radiation acting on a body it exerts pressure. So the radiation from the core of the collapsing matter/dust will exert a force on the collapsing dust on the surface of the star which acts opposite to the gravitational force directed towards the center. If the radiation is too large, this outward radiation pressure will blow away the collapsing matter and thus the star will not form. Thus we end up with an upper limit on the brightness of the stars, which is known as the *Eddington limit*. This quantity is intimately connected to the cross-section which is given in terms of the ratio of the radiated power by the particles and the incident power per unit area (flux of the power) on the particle. As done in problem 2, we again consider a charge q placed on an electromagnetic wave of amplitude E . (Let us employ Gaussian units in the following and ignore certain factors, which will not affect our conclusion. So, only work with whatever quantities are given to you.)

(a) What is the acceleration of the particle?

The power radiated by a charge undergoing an acceleration a is given by $P \approx q^2 a^2 / c^3$. The flux of the power incident is given by the Poynting vector which is expressed by $S = cE^2 / 4\pi$. The cross-section is given by $\sigma = P/S$.

(b) Find the final expression for σ in terms of the given quantities.

(c) Evaluate the value of σ for protons.

(d) If the flux density of photons on the stars surface is n , roughly determine that in terms of the luminosity of the star, L . Note that luminosity is the energy radiated by the star per unit time. Assume that the frequency of the radiation is just ω .

(e) Find the outward force exerted by the radiation on the infalling proton (Hint: think of Newton's law and the de Broglie relation which is also valid for photons).

(f) Now this force must not exceed the gravitational force on the proton , otherwise the proton gets blown away. From this find the upper limit on the Luminosity of the star. Can you produce a number? (The mass of our Sun is 2×10^{30} kg).

0.5+1+0.5+1+2+3 =8